



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 7.5

Obtained Marks: _____

Object Orientation Programming (LAB)

Project Proposal

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Hotel Management System

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1. Introduction

The Hotel Management System (HMS) is a software designed to simplify hotel operations, making them more efficient and improving customer service. It uses Object-Oriented Programming (OOP) to handle tasks like room bookings, check-outs, and payments through an easy-to-use interface. By organizing room details and hotel processes into structured components, the system is easy to update and expand.

HMS helps hotel managers by reducing mistakes, improving room assignments, and enhancing customer service. It provides real-time room availability, automates booking and check-out, and keeps guest information secure.

This report explains the system's features, how it works, and possible future improvements. It highlights the benefits of HMS, such as better efficiency, happier customers, and increased profits.

HMS is a useful tool for hotel managers looking to modernize their business. Its flexibility allows it to work well for hotels of all sizes.

2. Main Features of Project

- **Room Booking:**

Guests can view available rooms and book them by providing their details (Name, CNIC, and Contact Number).

- **Check-Out Functionality:**

Enables guests to check out by entering their name, freeing the room for future bookings.

- **Payment Processing:**

Supports multiple payment methods, including bank transfers and mobile payment services, making it convenient for guests.

- **User Authentication:**

Secures access to the system with a password, ensuring that only authorized personnel can manage hotel operations.

- **Guest Information Management:**

Stores and displays guest information securely and in real time, allowing staff to retrieve data easily when needed.

- **Feedback Mechanism:**

Provides contact information for guests to submit feedback or inquiries, promoting customer engagement and service improvement.

- **Error Handling:**

Includes error handling for invalid user input, for example choosing an invalid option number from the menu or entering a character instead of an integer.

3. Background Information

Hotels and hospitality businesses rely on manual or outdated systems, which often lead to inefficiencies such as overbooking, lost reservations, and billing errors. Manual systems also pose threats to the guest information as anyone can access the details. A digital Hotel Management System helps overcome these challenges by automating operations and providing real-time information, as well as protects guest information, ensuring smooth hotel operations and better guest experiences.

4. Objectives

The main objectives of this project are:

- a. To develop a user-friendly hotel management software that automates reservations, billing, and guest management.
- b. To enhance operational efficiency by reducing human errors in manual booking and billing systems.
- c. To provide hotel administrators with a centralized dashboard for managing hotel activities.
- d. To improve customer satisfaction by providing a seamless booking experience and efficient hotel services.

5. Purpose

The purpose of the Hotel Management System is to:

- a. Enable hotel staff to manage bookings and room availability efficiently.
- b. Provide a smooth experience for customers while booking rooms and making payments.
- c. Allow hotels to collect guest feedback to improve services.

6. Scope

The scope of this project includes:

- Room booking and management.
- Guest information storage and retrieval.
- Payment processing.
- Basic reporting on room occupancy.
- Feedback management.

7. Limitations

- a. The system does not support basic financial tracking.
- b. It will not include real-time chat support for customers.
- c. Multi-language support is not included in the initial version.
- d. It does not include a front end in the initial version.

8. Tools and Techniques:

Tools:

- Programming Language: C++

Techniques:

- **Object-Oriented Programming (OOP):**

Application of classes, objects, encapsulation, inheritance, and polymorphism in a practical scenario.

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- **Software Development Methodologies:**

Understanding the Software Development Life Cycle (SDLC), including requirement analysis, design, implementation, testing, and maintenance.

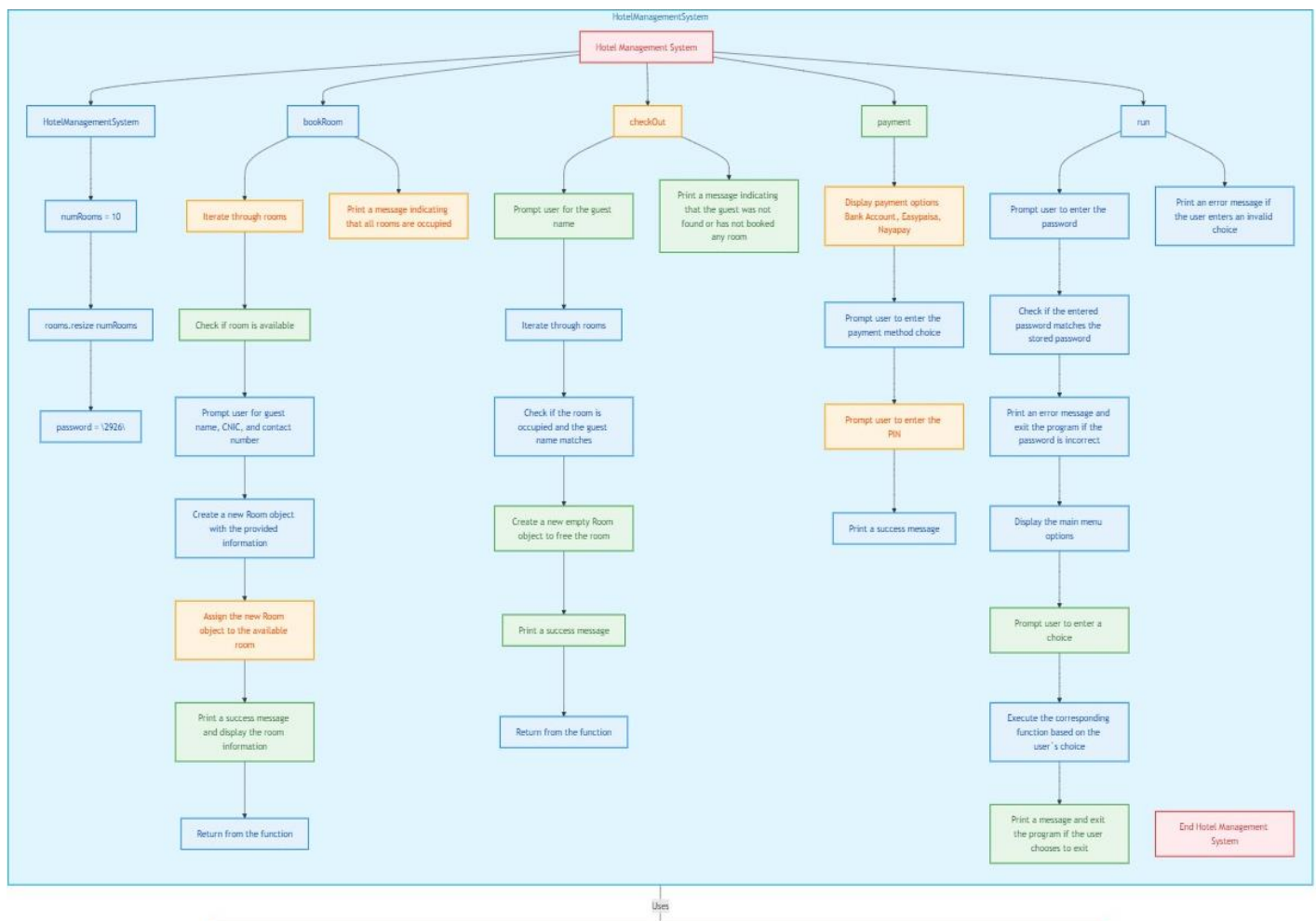
- **File Handling:**

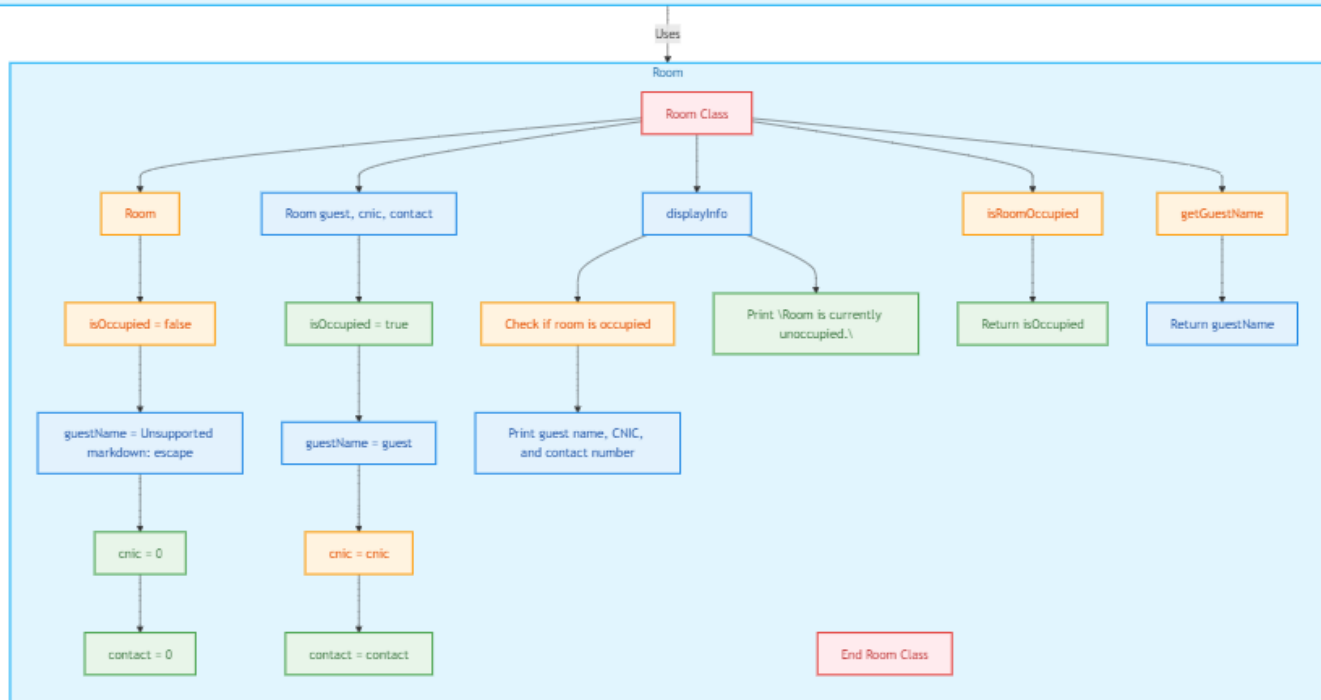
File handling and database management for storing customer and booking details..

- **Error Handling:**

Implementation of checks for user inputs to enhance the system's robustness and prevent invalid operations.

9. Flow Chart





10. Visual Aids:

- **Password**

```
Enter Password: 2926

"Welcome To Hotel Management System"

1. Book a Room
2. Check Out
3. For Feedback
4. Payment Method
5. Payment with Amount
0. Exit

Enter your choice: 1

Room 1 is available.
```

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- **Book Room and Check-out:**

```
Enter guest name to book: aa
Enter CNIC: 11
Enter Contact No: 11
Room 1 booked for aa.
Name: aa
CNIC: 11
Contact No: 11

"Welcome To Hotel Management System"

1. Book a Room
2. Check Out
3. For Feedback
4. Payment Method
5. Payment with Amount
0. Exit
Enter your choice: 2
Enter guest name to check out: aa
Guest checked out of Room 1.
```

- **Payment Method**

```
"Welcome To Hotel Management System"

1. Book a Room
2. Check Out
3. For Feedback
4. Payment Method
5. Payment with Amount
0. Exit
Enter your choice: 4
1. Bank Account
2. Easypaisa
3. Nayapay
1
Enter PIN
1021
Payment Done
Thank you
```

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11. Conclusion

The Hotel Management System demonstrates the effective application of OOP principles to solve real-world problems in the hospitality industry. By automating key processes, the system significantly enhances operational efficiency and improves customer satisfaction. Future developments could include expanding payment options, adding a graphical user interface (GUI), and implementing features for reporting and analytics. This project not only fulfills academic requirements but also serves as a valuable learning experience in software development and system design.

12. References

- a. C++ Programming Guide [Reference Books & Online Resources]
- b. Database Management in Hospitality [Research Articles]
- c. Chat GPT

13. Work Disribution

Aiman Nasir	User Authentication and Main Menu
Tehniat Khan	Booking and Check-out
Syeda Nalain Zainab Bukhari	Main Function
Malika Naz	Feedback and Payment System
Safia Alam	File Handling